

Test 1 · Algebra

Grade 8

40 minutes 100 marks 14 questions + bonus No calculator

Lessons 1–8 · Revision of Grade 7 · Algebraic expressions · Cancelling algebraic fractions

- Show your working — method carries marks.
- Whenever you cancel a fraction, state the permissible values of the variable.

NAME

CLASS

DATE

MARK / 100

A

Section A · Short answers

EASY 30 marks

Five marks each. One line of working is enough.

- 1

Simplify: $x^6 \cdot x^4$
- 2

Expand: $4(3x - 2)$
- 3

Factorise: $10a - 15$
- 4

State the permissible values of $\frac{1}{x - 6}$
- 5

Find the value of $\frac{x + 5}{x - 2}$ when $x = 3$
- 6

Cancel: $\frac{15x^3}{5x}$

B

Section B · Eight marks each

MEDIUM 40 marks

Method marks are available even when the final answer is wrong.

- 7

Simplify: $(3a^3b^2)^2$ [8]
- 8

Factorise: $9x^2 - 16y^2$ [8]
- 9

State the permissible values of $\frac{x + 2}{x^2 - 25}$ [8]
- 10

For which value of x is $\frac{x - 7}{x + 1}$ equal to zero? [8]

11 Cancel: $\frac{3x + 12}{x^2 - 16}$ [8]

C Section C · Ten marks each

HARD 30 marks

Factorise before you cancel, and state the values the variable may not take.

12 (a) Factorise completely: $x^3 - 9x$ [4]

(b) Hence cancel $\frac{x^3 - 9x}{x^2 + 3x}$ and state the permissible values. [6]

13 Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$, state the permissible values, then find its value when $x = 9$. [10]

14 (a) Cancel: $\frac{2x^2 - 2y^2}{3y - 3x}$ [7]

(b) Explain in one sentence why $\frac{x + 5}{5}$ cannot be cancelled. [3]

Bonus

★ Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS +10 marks

15 (a) Show that $\frac{m^2 - mn}{n^2 - mn} = -\frac{m}{n}$. [7]

(b) State the permissible values. [3]